PMA Note 1

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1 The Real and Complex Number Systems

1.1 Introduction

Example 1.1. It is well known that there is no rational p such that $p^2 = 2$.

Define $A \coloneqq \left\{p \in \mathbb{Q}^+ \mid p^2 < 2\right\}$ and $B \coloneqq \left\{p \in \mathbb{Q}^+ \mid p^2 > 2\right\}$. Then A contains no largest number and B contains no smallest.

Proof. It suffices to show that

$$\forall p \in A,\ \exists q \in A\ s.t.\ p < q$$

and

$$\forall p \in B,\ \exists q \in B\ s.t.\ q < p.$$

For every $p \in \mathbb{Q}^+$, take

$$q = p - \frac{p^2 - 2}{p + 2} = \frac{2p + 2}{p + 2}. \tag{1}$$

Then

$$q^2 - 2 = \frac{2(p^2 - 2)}{(p + 2)^2}. \tag{2}$$

Hence,

$$\begin{aligned} p \in A &\implies p^2 - 2 < 0 \\ &\implies p < q \text{ and } q^2 < 2 && \because (1), (2) \\ &\implies q \in A \end{aligned}$$

and

$$\begin{aligned} p \in B &\implies p^2 - 2 > 0 \\ &\implies 0 < q < p \text{ and } q^2 > 2 && \because (1), (2) \\ &\implies q \in B. \end{aligned}$$

We have shown the desired result. □

1.2 Ordered Sets

Definition 1.2. Let S be a set. An *order* on S is a relation, denoted by $<$, with the following two properties:

(i) If $x \in S$ and $y \in S$ then one and only one of the statements

$$x < y, \qquad x = y, \qquad y < x$$

is true.

(ii) If $x, y, z \in S$, if $x < y$ and $y < z$, then $x < z$.

Definition 1.3. An *ordered set* is a set S in which an order is defined.

Definition 1.4. Suppose S is an ordered set, and $E \subset S$. If there exists a $\beta \in S$ such that $x \leq \beta$ for every $x \in E$, we say that E is *bounded above*, and call β an *upper bound* of E.

Lower bounds are defined in the same way (with $\geq$ in place of $\leq$).

Definition 1.5. Suppose S is an ordered set, $E \subset S$, and E is bounded above. Suppose there exists an $\alpha \in S$ with the following properties:

(i) α is an upper bound of E.

(ii) If $\gamma < \alpha$ then γ is not an upper bound of E.

Then α is called the *least upper bound of* E [that there is at most one such α is clear from (ii)] or the *supremum of* E, and we write

$$\alpha = \sup E.$$

The *greatest lower bound*, or *infimum*, of a set E which is bounded below is defined in the same manner: The statement

$$\alpha = \inf E$$

means that α is a lower bound of E and that no β with $\beta > \alpha$ is a lower bound of E.

Definition 1.6. An ordered set S is said to have the *least-upper-bound property* if the following is true:

If $E \subset S$, E is not empty, and E is bounded above, then $\sup E$ exists in S.

Remark 1.7. The ordered set $\mathbb{Q}$ does not have the least-upper-bound property.

The following theorem shows that every ordered set with the least-upper-bound property also has the greatest-lower-bound property.

Theorem 1.8. *Suppose S is an ordered set with the least-upper-bound property, $B \subset S$, B is not empty, and B is bounded below. Let L be the set of all lower bounds of B. Then*

$$\alpha = \sup L$$

exists in S, and $\alpha = \inf B$.

In particular, $\inf B$ *exists in S.*

Proof. Since B is bounded below, L is nonempty. Let $x \in B$. Then $y \leq x$ for every $y \in L$. Thus, every $x \in B$ is an upper bound of L. Since L is a nonempty subset of S that is bounded above, by hypothesis,

$$\alpha = \sup L$$

exists in S.

If $\gamma < \alpha$ then γ is not an upper bound of L, hence $\gamma \notin B$. It follows that $\alpha \leq x$ for every $x \in B$. Thus, $\alpha \in L$.

If $\alpha < \beta$ then $\beta \notin L$, since α is an upper bound of L.

We conclude that α is a lower bound of B (since $\alpha \in L$) and if $\alpha < \beta$ then β is not a lower bound of B (since $\beta \notin L$), which implies $\alpha = \inf B$. $\square$

Remark 1.9. Fix $x \in B$. Then $y \leq x$ for all $y \in L$. Thus, $\sup L \leq x$. Since the choice of $x \in B$ was arbitrary, we have

$$\sup L = \alpha \leq x \text{ for every } x \in B,$$

which implies α is a lower bound of B. Hence, $\alpha \in L$.

1.3 Fields

Definition 1.10. A *field* is a set F with two operations, called *addition* and *multiplication*, which satisfy the following so-called "field axioms" (A), (M), and (D):

(A) Axioms for addition

- (A1) If $x \in F$ and $y \in F$, then their sum $x + y$ is in F.
- (A2) Addition is commutative: $x + y = y + x$ for all $x, y \in F$.
- (A3) Addition is associative: $(x + y) + z = x + (y + z)$ for all $x, y, z \in F$.
- (A4) F contains an element 0 such that $0 + x = x$ for every $x \in F$.
- (A5) To every $x \in F$ corresponds an element $-x \in F$ such that

$$x + (-x) = 0.$$

(M) Axioms for multiplication

- (M1) If $x \in F$ and $y \in F$, then their product xy is in F.
- (M2) Multiplication is commutative: $xy = yx$ for all $x, y \in F$.
- (M3) Multiplication is associative: $(xy)z = x(yz)$ for all $x, y, z \in F$.
- (M4) F contains an element $1 \neq 0$ such that $1x = x$ for every $x \in F$.
- (M5) If $x \in F$ and $x \neq 0$ then there exists an element $1/x \in F$ such that

$$x \cdot (1/x) = 1.$$

(D) The distributive law

$$x(y + z) = xy + xz$$

holds for all $x, y, z \in F$.

Definition 1.11. An *ordered field* is a *field* F which is also an *ordered set*, such that

(i) $x + y < x + z$ if $x, y, z \in F$ and $y < z$.

(ii) $xy > 0$ if $x \in F$, $y \in F$, $x > 0$, and $y > 0$.

1.4 The Real Field

Theorem 1.12. *There exists an ordered field $\mathbb{R}$ which has the least-upper-bound property. Moreover, $\mathbb{R}$ contains $\mathbb{Q}$ as a subfield.*

Theorem 1.13.

(a) If $x \in \mathbb{R}$, $y \in \mathbb{R}$, and $x > 0$, then there is a positive integer n such that

$$nx > y.$$

(b) If $x \in \mathbb{R}$, $y \in \mathbb{R}$, and $x < y$, then there exists a $p \in \mathbb{Q}$ such that $x < p < y$.

Proof.

(a) Let $x, y \in \mathbb{R}$ and $x > 0$. Suppose to the contrary that $nx \leq y$ for each $n \in \mathbb{N}$ and define

$$A := \{nx \mid n \in \mathbb{N}\}.$$

Then A is nonempty and bounded above by y. Thus, the supremum of A exists in $\mathbb{R}$. Put $\alpha = \sup A$.

Since $x > 0$, we have $\alpha - x < \alpha$, hence, $\alpha - x$ is not an upper bound of A. Therefore, $\alpha - x < mx$ for some $m \in \mathbb{N}$. Then $\alpha < (m + 1)x \in A$, which is a contradiction, since α is an upper bound of A.

(b) Since Rudin's proof of part (b) is already explicit, we omit it here. □

Theorem 1.14. *For every real $x > 0$ and every integer $n > 0$ there is one and only one positive real y such that $y^n = x$.*

This number y is written $\sqrt[n]{x}$ or $x^{1/n}$.

Proof. We omit the proof here. □

Corollary 1.15. *If a and b are positive and real numbers and n is a positive integer, then*

$$(ab)^{\frac{1}{n}} = a^{\frac{1}{n}} b^{\frac{1}{n}}.$$

Proof. We omit the proof here. □

Lemma 1.16.

(a) If $E \subset \mathbb{Z}$ has the infimum, then $\inf E \in E$.

(b) If $x \in \mathbb{R}$ then there exists an $n \in \mathbb{Z}$ such that

$$n \leq x < n + 1.$$

Proof.

(a) Let $\alpha = \inf E$. Suppose to the contrary that $\alpha \notin E$. Since $\alpha + 1$ is not a lower bound of E, choose an $x_0 \in E$ such that

$$\alpha < x_0 < \alpha + 1. \tag{3}$$

Since x_0 is not a lower bound of E, choose an $x_1 \in E$ again such that

$$\alpha < x_1 < x_0. \tag{4}$$

Subtracting x_0 from (4), we obtain $\alpha - x_0 < x_1 - x_0 < 0$. Since $-1 < \alpha - x_0$ from (3), it follows that $-1 < x_1 - x_0 < 0$. This contradicts the fact that $x_1 - x_0 \in \mathbb{Z}$. We conclude that $\alpha \in E$.

(b) Let $x \in \mathbb{R}$ and define $E \coloneqq \{n \in \mathbb{Z} \mid n > x\}$. By the Archimedean Property, E is nonempty. Since E is bounded below by x, the infimum of E exists in $\mathbb{R}$. Since $E \subset \mathbb{Z}$, $\inf E$ is the least element of E by (a). Thus, we have

$$\inf E - 1 \leq x < \inf E.$$

Putting $n = \inf E - 1 \in \mathbb{Z}$, we obtain the desired inequality. □

Remark 1.17 (Decimals). Let $x > 0$ be real. Let n_0 be the largest integer such that $n_0 \leq x$, by lemma 1.16. Having chosen $n_0, n_1, \ldots, n_{k-1}$, let n_k be the largest integer such that

$$n_0 + \frac{n_1}{10} + \cdots + \frac{n_k}{10^k} \leq x.$$

Let E be the set of these numbers

$$n_0 + \frac{n_1}{10} + \cdots + \frac{n_k}{10^k} \qquad (k = 0, 1, 2, \ldots). \tag{5}$$

Notice that x is an upper bound of E and if $k \in \mathbb{N}$ then

$$n_0 + \frac{n_1}{10} + \cdots + \frac{n_k}{10^k} \leq x < n_0 + \frac{n_1}{10} + \cdots + \frac{n_k + 1}{10^k}.$$

Hence, we have

$$x - n_0 - \frac{n_1}{10} - \cdots - \frac{n_k}{10^k} < \frac{1}{10^k} \text{ for } k \in \mathbb{N}. \tag{6}$$

Suppose $y < x$. Then by the Archimedean Property, there exists a positive integer k such that $1/10^k < x - y$. From (6), we obtain

$$x - n_0 - \frac{n_1}{10} - \cdots - \frac{n_k}{10^k} < x - y,$$

hence

$$y < n_0 + \frac{n_1}{10} + \cdots + \frac{n_k}{10^k}.$$

Therefore, $x = \sup E$. The decimal expansion of x is

$$n_0 \cdot n_1 n_2 n_3 \cdots . \tag{7}$$

Conversely, for any infinite decimal (7), the set E of numbers (5) is bounded above, and (7) is the decimal expansion of $\sup E$.

Exercises

Exercise 1. Let A be a nonempty set of real numbers which is bounded below. Let $-A$ be the set of all numbers $-x$, where $x \in A$. Prove that

$$\inf A = -\sup(-A).$$

Proof. Since A is bounded below, the infimum of A exists in $\mathbb{R}$. If $a \in A$ then $\inf A \leq a$. Hence,

$$-a \leq -\inf A \text{ for all } a \in A, \tag{8}$$

and it follows that $-A$ is bounded above. Thus, $\sup(-A)$ exists in $\mathbb{R}$. Since $-\inf A$ is an upper bound of $-A$ from (8), we have $\sup(-A) \leq -\inf A$, hence

$$\inf A \leq -\sup(-A). \tag{9}$$

Since $-a \leq \sup(-A)$ for every $a \in A$, we obtain $-\sup(-A) \leq a$ for all $a \in A$. Therefore, $-\sup(-A)$ is a lower bound of A and

$$-\sup(-A) \leq \inf A. \tag{10}$$

Combining (9) and (10), we conclude that $\inf A = -\sup(-A)$. $\square$

Exercise 2. Fix $b > 1$.

(a) If m, n, p, q are integers, $n > 0$, $q > 0$, and $r = m/n = p/q$, prove that

$$(b^m)^{1/n} = (b^p)^{1/q}.$$

Hence it makes sense to define $b^r = (b^m)^{1/n}$.

(b) Prove that $b^{r+s} = b^r b^s$ if r and s are rational.

(c) If x is real, define $B(x)$ to be the set of all numbers b^t, where t is rational and $t \leq x$. Prove that

$$b^r = \sup B(r)$$

when r is rational. Hence it makes sense to define

$$b^x = \sup B(x)$$

for every real x.

(d) Prove that $b^{x+y} = b^x b^y$ for all real x and y.

Proof.

(a) Let $b^{mq} = b^{np} = x$. Since n and q are positive integers and $b > 0$, by Theorem 1.14, we have

$$b^m = x^{1/q} \text{ and } b^p = x^{1/n}.$$

It follows that

$$(b^m)^{1/n} = (x^{1/q})^{1/n} = (x^{1/n})^{1/q} = (b^p)^{1/q},$$

since $[(x^{1/q})^{1/n}]^{nq} = [(x^{1/n})^{1/q}]^{nq} = x$.

(b) Let m, n, p, q be integers and $n > 0$, $q > 0$ and let $r = m/n$ and $s = p/q$. Then by Corollary 1.15, we obtain

$$\begin{aligned} b^{r+s} &= b^{\frac{m}{n}+\frac{p}{q}} = b^{\frac{mq+np}{nq}} \\ &= \left(b^{mq+np}\right)^{\frac{1}{nq}} \\ &= \left(b^{mq}b^{np}\right)^{\frac{1}{nq}} \\ &= \left(b^{mq}\right)^{\frac{1}{nq}} \left(b^{np}\right)^{\frac{1}{nq}} \\ &= b^{\frac{mq}{nq}} b^{\frac{np}{nq}} = b^{\frac{m}{n}} b^{\frac{p}{q}} = b^r b^s. \end{aligned}$$

(c) Note that by Theorem 1.14, $b^r > 0$ if $b > 1$ and r is rational.

Let n be a positive integer and let $0 < a < 1$ be real. We first show that $0 < a^{1/n} < 1$. Suppose to the contrary that $a^{1/n} \geq 1$. Then we obtain

$$1 \leq \left(a^{1/n}\right)^n = a < 1,$$

a contradiction. Therefore, if $0 < a < 1$ then $0 < a^{1/n} < 1$, where n is a positive integer.

Suppose r and s are rational. Let m, n, p, and q be integers and let n and q be positive. Then we write $r = m/n$, $s = p/q$.

Suppose $r < s$. Then

$$r - s = \frac{m}{n} - \frac{p}{q} = \frac{mq - np}{nq} < 0,$$

hence

$$0 < b^{mq-np} = \frac{1}{b^{np-mq}} < 1$$

and

$$0 < b^{r-s} = b^{\frac{mq-np}{nq}} = \left(b^{mq-np}\right)^{1/nq} < 1.$$

Thus, if $r < s$ then

$$b^r = b^{(r-s)+s} = b^{r-s}b^s < b^s.$$

If t is rational and $t \leq r$ then $b^t \leq b^r$. It follows that $B(r)$ is bounded above by b^r and $\sup B(r)$ exists in $\mathbb{R}$. Since b^r is an upper bound of $B(r)$,

$$\sup B(r) \leq b^r.$$

Since $b^r \in B(r)$,

$$b^r \leq \sup B(r).$$

We conclude that $b^r = \sup B(r)$.

(d) Let t_1 and t_2 be rational. If $t_1 \leq x$ and $t_2 \leq y$ then $t_1 + t_2 \leq x + y$. Thus, we have

$$b^{t_1} b^{t_2} = b^{t_1 + t_2} \leq \sup B(x + y) = b^{x+y}.$$

Fix $t_1 \leq x$. Then

$$b^{t_2} \leq b^{x+y}/b^{t_1} \text{ for all } t_2 \leq y.$$

Hence,

$$b^y = \sup B(y) \leq b^{x+y}/b^{t_1}.$$

Since the choice of t_1 was arbitrary,

$$b^{t_1} \leq b^{x+y}/b^y \text{ for all } t_1 \leq x,$$

and it follows that

$$b^x = \sup B(x) \leq b^{x+y}/b^y.$$

Therefore, we obtain

$$b^x b^y \leq b^{x+y}. \tag{11}$$

Notice that

$$0 < b^{t_1} \leq b^x \text{ for every } t_1 \leq x$$

and

$$0 < b^{t_2} \leq b^y \text{ for every } t_2 \leq y.$$

Hence, for every $t_1 \leq x$ and $t_2 \leq y$,

$$b^{t_1 + t_2} = b^{t_1} b^{t_2} \leq b^x b^y,$$

and it follows that

$$b^{x+y} = \sup B(x + y) \leq b^x b^y. \tag{12}$$

Combining (11) and (12), we conclude that

$$b^{x+y} = b^x b^y.$$

□

Exercise 3. Prove that no order can be defined in the complex field that turns it into an ordered field.

Proof. Suppose to the contrary that there exists an order $<$ on $\mathbb{C}$ which turns $\mathbb{C}$ into an ordered field. Let 0 and 1 denote the additive and multiplicative identities, respectively. Then $0 = (0, 0)$ and $1 = (1, 0)$.

Note that $i = (0, 1) \neq 0$. If $x \neq 0$ then $x^2 > 0$. It follows that

$$i^2 = (0, 1)(0, 1) = (-1, 0) > 0.$$

Since $(-1, 0) + (1, 0) = 0$, $(-1, 0)$ is the additive inverse of $(1, 0)$. Since $1 > 0$ and if $x > 0$ then $-x < 0$ in any ordered field, we obtain

$$(-1, 0) = -(1, 0) = -1 < 0.$$

This contradicts the Trichotomy Property. □

2 Basic Topology

2.1 Finite, Countable, and Uncountable Sets